The Hidden Factory: Why Manufacturers Stop Short of Their Real Capacity

A white paper on Zero Loss Analysis

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Executive summary

Most plants run to a perceived ceiling they believe is physical. In many cases it is not. It is the point at which a team stopped investigating a hard problem at the bottleneck, which then got written into the standards and the accounting system and has been defended ever since.

This paper makes five points.

- **Plants set their own limits.** A speed that "can't be exceeded" is usually the last speed at which a problem did not appear, not the true limit of the equipment.
- **The limit is rarely challenged.** Once it is written into standard rates, budgets and efficiency targets, the ceiling stops being a hypothesis and becomes the baseline.
- **Measures like OEE conceal the gap.** They report performance against the standard, so a plant can look excellent against a target that was set too low.
- **The problem can be solved with a structured method.** Variance Tree Analysis, one of the tools within Zero Loss Analysis, breaks a problem into measurable factors and isolates the real cause.
- **The financial consequence is large.** Recovered capacity is capacity the plant has already paid for, and it can defer or avoid capital spent on new lines. We call this the hidden plant.

The rest of this paper explains how ceilings get built, how to take them apart, and how continuous improvement teams can learn to do it themselves.

The limit that isn't

Every manufacturing plant has a number it believes is the ceiling. Ask the line manager what the filler can run at, and they won't hesitate — they'll give you a figure to the bottle per minute. Ask how they know, and the answer is always some version of the same story: we tried to go faster, and it broke.

That number gets written into the standard. It gets built into the accounting system as the basis for efficiency reporting. It becomes the ceiling the plant measures itself against, year after year, in every OEE report and every monthly ops review.

The problem is that the ceiling was never actually tested. It was abandoned.

How a ceiling gets built

Here is the pattern, almost without exception, across every plant we've examined.

A team tries to speed up a line. Somewhere past a certain speed, a defect appears — in this case, a beer filler that produced low fills every time the line ran above a certain speed. The team tries a few obvious fixes. Nothing works. The problem gets labeled as a physical limit of the equipment, and the line gets dialed back to the speed where the defect goes away.

From that day forward, that speed is the plant's "maximum capability." It goes into the standard. The accounting system measures efficiency against it. Every CI initiative from then on is measured against that number, not against what the equipment could actually do.

Nobody is lying. Nobody is hiding anything. The team that gave up on the problem gave up because they ran out of obvious things to try — not because the problem was unsolvable. But once a limit is written into the accounting system, it stops being a hypothesis and starts being treated as physics.

Where the ceiling gets locked in

A ceiling that lives only in people's heads can still be challenged. A new engineer joins, asks why the filler never runs faster, and someone explains the history. The ceiling becomes much harder to move when it leaves people's heads and enters the systems the plant runs on.

In most plants that happens in four places.

Standards. The rate the line "can" run becomes the standard rate in the planning, ERP or manufacturing execution system. Production schedules, labor plans and delivery commitments are all built on it.

Costing. Standard cost per unit is calculated from the standard rate, and variances are reported against it. A line that hits its standard is on target, whatever it might have been able to do.

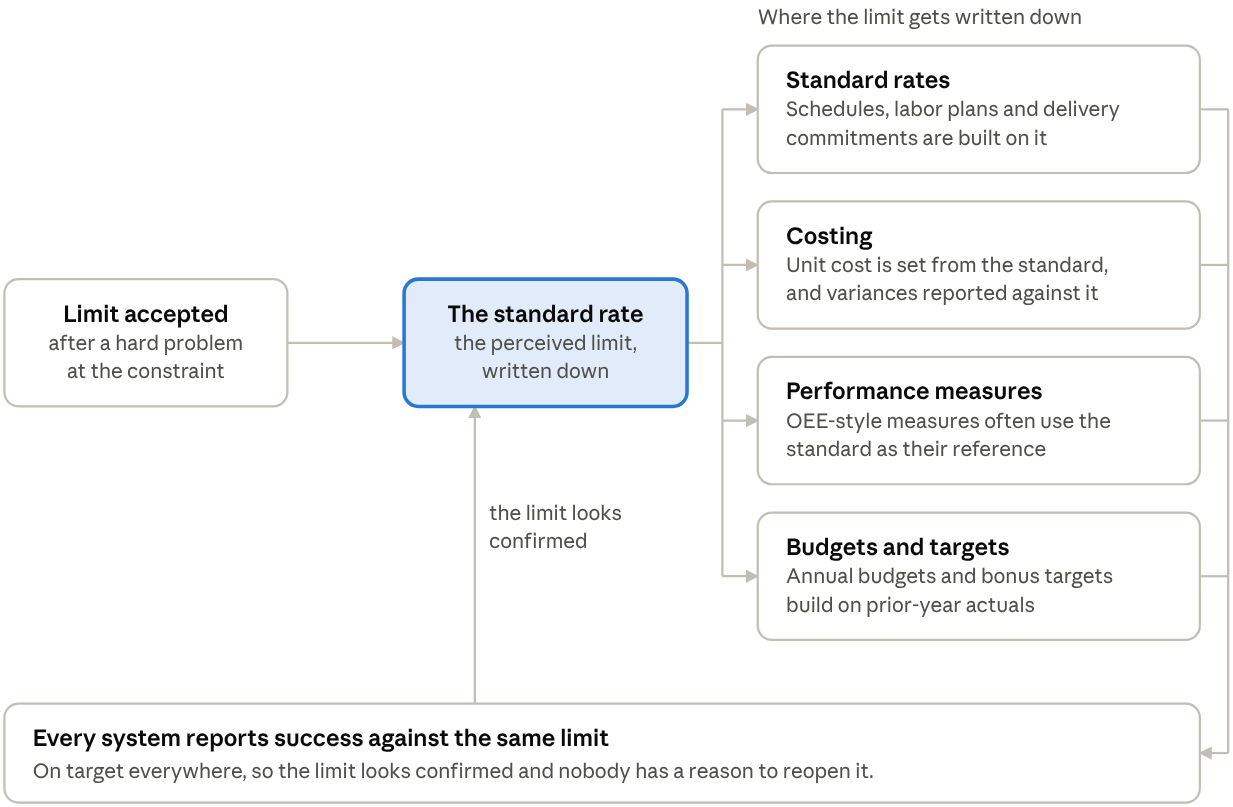
Performance measures. In many plants OEE and similar metrics use the standard rate as the reference for the performance component. A line running at the standard reads as 100 percent on that component.

Budgets and targets. Annual budgets, bonus targets and improvement goals are typically set as increments on last year's actual results, which were themselves produced by running to the standard.

Once the same number sits in all four places, challenging it means challenging the schedule, the cost model, the scorecard and the budget at the same time. Nobody is rewarded for doing that, and plenty of people have a reason not to. The rational response is to leave it alone.

This is not a failure of the people involved. It is how organizations behave when a number becomes load-bearing. The result is a limit that is institutional rather than physical, and that is invisible to the very people who are measured against it.

One accepted limit feeds four systems, and each reports the plant on target



how an accepted limit flows into four systems · 4 systems, 1 feedback loop

Each system reinforces the others. A line that runs at the standard is on target in all four, so no report ever suggests that the standard itself might be wrong.

The fill tubes

On that beer line, the low-fill problem looked exactly like an unsolvable equipment limit. Every time the line sped up past the threshold, fills came in short. The team had already tried the obvious adjustments. The conclusion was that the filler simply couldn't run faster without compromising fill quality — so the plant settled on the slower speed as its ceiling.

We approached it differently. Instead of guessing at causes, we used a tool we call Variance Tree Analysis — a structured, scientific way of breaking a problem down into every factor that could plausibly contribute to it, then measuring each one to find out which factor is actually responsible. It replaces brainstorming with systematic elimination.

The variance tree pointed to something nobody had looked at: the fill tubes were the wrong diameter. At higher line speeds, beer simply couldn't flow into the bottle fast enough through tubes that were too narrow — hence the low fills. It wasn't a limit of the filler. It was a size of pipe.

Once the tubes were resized, the line ran faster with no fill problem at all. The "physical limit" the plant had built into its standards for years turned out to be an undersized component.

What the fill tube case teaches

The result: a 20% increase in output.

Three lessons from this case apply well beyond breweries.

The observation was right and the conclusion was wrong. Low fills really did appear when the line sped up. The team was not imagining the problem. Where they went wrong was in the inference: "we get low fills above this speed" became "this is the fastest the filler can run." Those are different statements, and only the first one had been established.

Diagnosis beat opinion. The team could have gone on debating causes indefinitely, and a debate between experienced people rarely settles anything. Measuring each factor in turn converted an argument into a finding.

The answer was a change to one component, not a new line. The cause turned out to be a part that could be changed. That is the usual shape of these findings: a specific, physical, fixable cause that nobody had isolated, sitting behind a limit everybody had accepted.

Why this keeps happening

This is not a one-off story. It is the normal life cycle of a bottleneck in almost every plant:

A team hits a hard problem at the constraint. They try the fixes that are obvious to people without a structured method for decomposing the problem. The fixes don't work, because the obvious fixes are rarely the actual cause. The team concludes the limit is physical and permanent. The plant writes that limit into its standards, its accounting, and its OEE targets.

From that point on, the plant is not measuring itself against what the equipment can do. It is measuring itself against what a team, on a particular day, with the tools they had, was able to figure out. Those are very different numbers — and the gap between them is what

we call the hidden factory: capacity the plant already owns, has already paid for, and isn't using, because nobody has gone back to re-open a question that was closed years ago.

Why plants chase the visible losses

Most plants put their improvement effort into the losses they can see. That is understandable, and it explains a good deal of why the hidden factory stays hidden.

Visible losses announce themselves. A stop has a start time, a duration, a fault code and a work order. It lands on a Pareto chart, and the chart tells a team where to start. A chronic limit does nothing of the kind. A line that runs steadily at a speed it could exceed produces no alarm and no record.

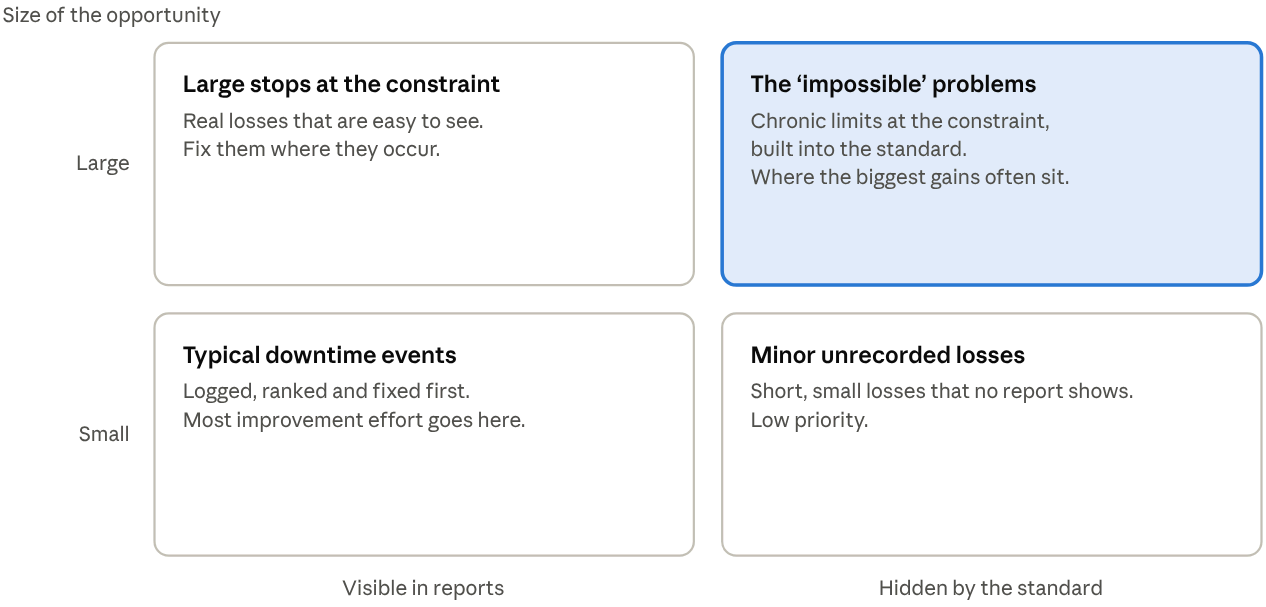
The measurement system rewards what it can see. Downtime is captured, ranked and reviewed every week. The standard rate is rarely questioned, so the losses behind it never appear in a report.

A small win is safer than a large uncertain one. A breakdown has an owner and an end point, and the ticket closes within days. A problem that has already defeated the team once has no owner, an uncertain outcome and a visible history of failure. Teams judged on results choose the sure win.

The usual tools suit events, not chronic losses. Root cause analysis works well on a breakdown because there is an event to investigate. A chronic loss has no trigger, so brainstorming returns the explanations the team already believes.

Reopening an old problem carries a social cost. "We tried that" is often true, and some of the people who tried are still there. Leaving the limit alone is the safer choice.

Effort goes to visible losses; the biggest gains often sit in hidden ones



Illustrative. Positions show typical patterns, not measurements.

visible versus hidden losses · 2x2, illustrative

None of this means visible losses should be ignored. Where a plant has large stops at its constraint, they should be fixed. The point is that visible losses are often not the largest ones, and the ranking is hard to see because of how plants measure. A chronic limit at the constraint affects output every minute the line runs, while downtime at a resource that is not the constraint has little effect on the output of the plant.

Why OEE hides the problem instead of revealing it

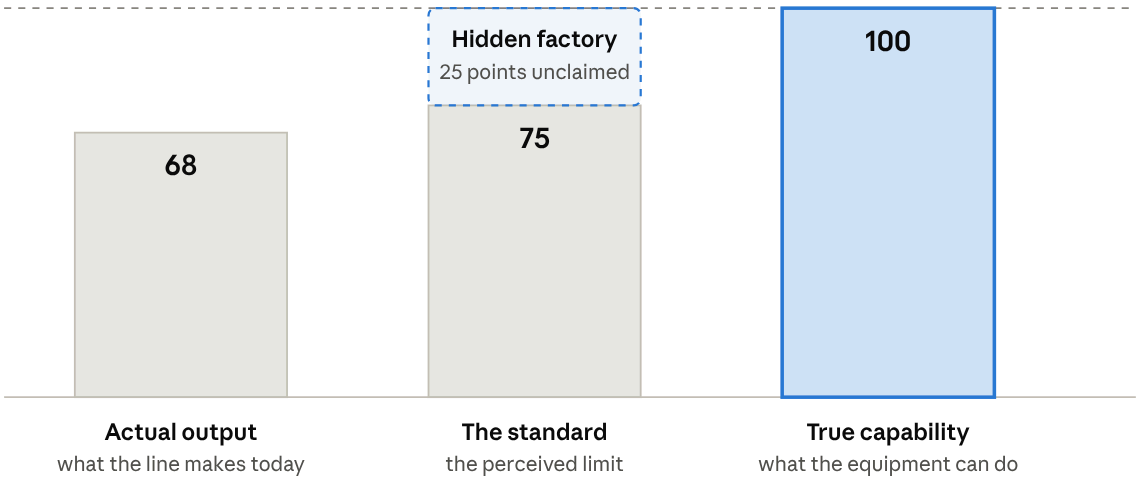
OEE and similar efficiency measures are built to compare current performance against a standard. That is exactly what makes them blind to this problem.

If the standard itself was set at the ceiling of what one team could figure out — rather than at the true capability of the equipment — then OEE will cheerfully report 95% efficiency against a number that is already 20 to 30% below what the line could really do. The measurement system is internally consistent and completely wrong. It cannot see the gap because the gap was built into its own reference point.

This is why continuous improvement programs so often plateau. They optimize hard against a target that was never actually the plant's capability — it was just the last unsolved problem a team walked away from.

The standard sits below what the equipment can actually do

Output is 91% of the standard but only 68% of true capability



Illustrative figures, indexed so that true capability = 100. Not plant data.

illustrative example · index, true capability = 100

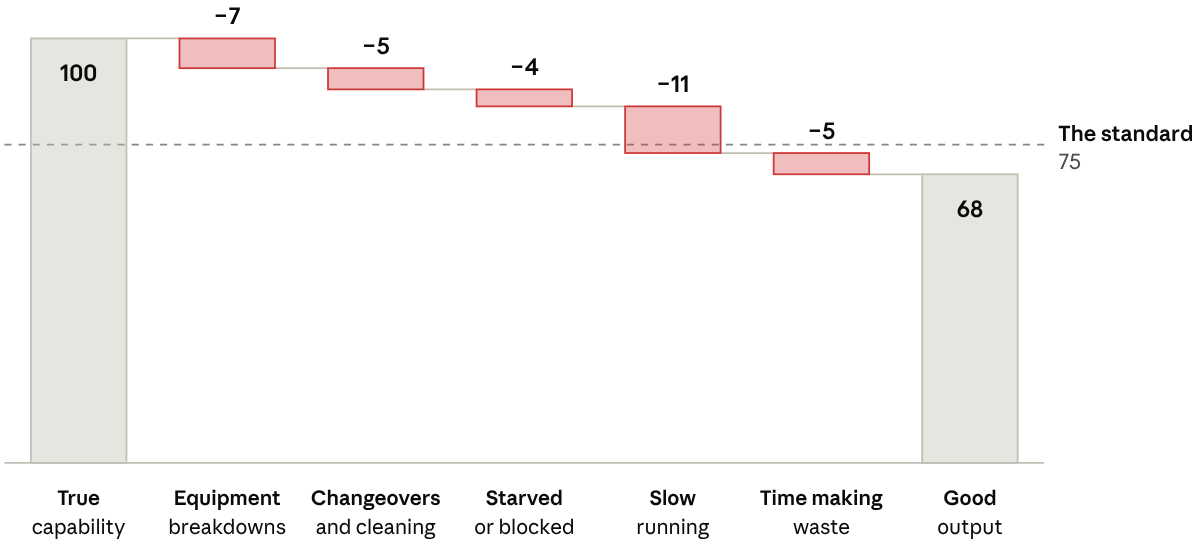
The report looks healthy because it measures against the wrong reference. The plant is close to its own standard, so nothing flags a problem, even though the standard itself sits well below what the equipment can do. The figures are illustrative and are not drawn from a specific plant.

Where the lost capacity goes

Zero Loss Analysis starts by accounting for the time at the constraint. Time that could have produced good output is lost in a small number of ways: downtime of various kinds, slow running, and time spent making product that has to be scrapped or reworked. Historical records seldom separate these categories cleanly, which is why they are best observed directly at the line.

Three kinds of loss absorb about a third of what the constraint could produce

32 of 100 points lost, leaving 68 as good output
Downtime 16 · Slow running 11 · Waste 5



Red steps are losses, grey bars are totals. Losses above the dashed line are already built into the standard. Illustrative figures in points of true capability. Not plant data.

illustrative example · points of true capability

Only the part of the loss that falls below the standard shows up in OEE. The rest was written into the standard when the limit was accepted, so it never appears as a loss at all. Recovering it is what Zero Loss Analysis is for.

Measuring against the right reference point

The answer is not to abandon OEE. It is to stop treating the standard as the end of the conversation. Two reference points matter, and a plant needs to see both.

The first is the standard: what the plant plans, costs and reports against. The second is the capability of the equipment: what the line could deliver if its known losses were removed. The distance between them is the hidden factory. Tracking that distance tells a plant something OEE cannot, which is how much of its own capacity it has not yet claimed.

Three habits follow from this.

Treat every limit as a hypothesis with a status. A limit at the constraint is either tested and confirmed, with evidence someone can show you, or inherited, meaning nobody can say how it was set. Inherited limits at the constraint are where to look first.

Reset the standard after a real improvement. If a gain is booked as a one-off favorable variance, it gets absorbed and forgotten. The plant then ends up with the same problem in reverse: a new capability that nobody plans around.

Report the gap, not just the score. An efficiency figure of 90 percent says little until it is clear what it is 90 percent of. Reporting the gap to demonstrated capability alongside it keeps the plant honest about what the number means.

What Variance Tree Analysis does differently

Variance Tree Analysis exists to re-open exactly these closed questions. Instead of asking a team to brainstorm causes — which tends to surface the same handful of familiar, comfortable explanations every time — it decomposes a performance gap into its full set of possible control factors, then tests each one against actual plant data until the true cause is isolated.

The fill tube problem is a representative example, not a special case. The same pattern shows up as a sizing problem, a control logic issue, a sequencing problem, a sensor placed in the wrong location — the specific cause varies by plant and by line, but the structure of the mistake is always the same: a team ran out of obvious ideas, and the plant mistook the edge of the team's ideas for the edge of the equipment's capability.

How a variance tree works

A variance tree starts with a gap. At the constraint there is a difference between what the equipment is delivering and what it should be able to deliver. The tree breaks that gap into the factors that together determine the result, and keeps breaking it down until each branch is something that can be measured and, if necessary, fixed.

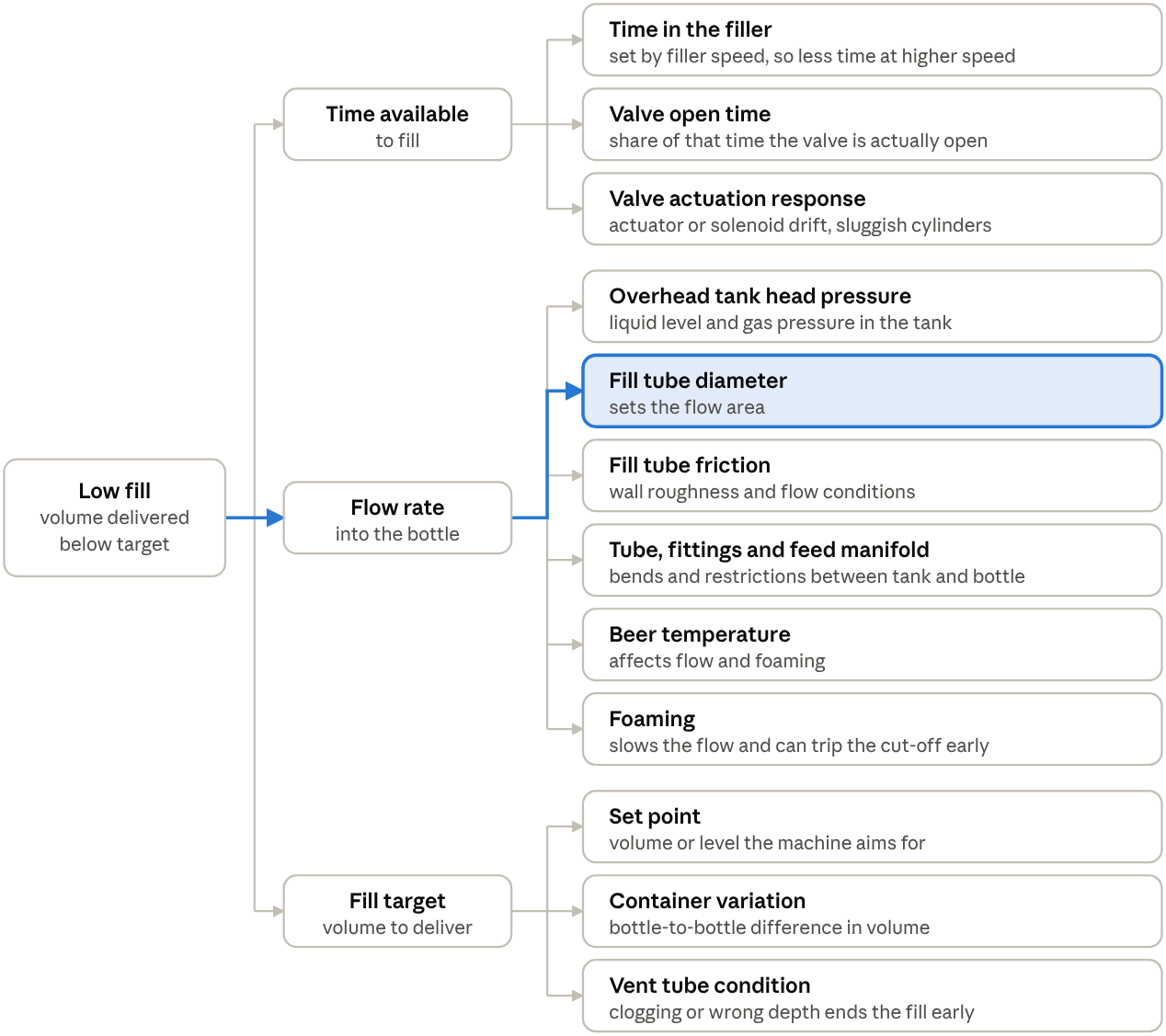
The method runs in seven steps.

1. **State the gap precisely.** Not "the filler is slow" but the specific difference between current and target output at the constraint, in units per hour.
2. **Break the gap into factors.** Each factor is a distinct contributor to the result. For a filler, examples are how long each fill takes, how much time is lost to stops, and how much product is rejected. Factors should be separate from one another, so that each branch can be assessed without arguing about the others.
3. **Define expected behavior for each factor.** What should this factor look like if the line were healthy?

4. **Measure actual behavior.** Observe the line or collect data for each factor, and compare it with what was expected.
5. **Eliminate what is behaving as expected.** Most branches will turn out to be fine. Closing them off is progress, not wasted effort, because it narrows the search with evidence.
6. **Go deeper on the factors that are not.** Break each of those into sub-factors and repeat until you reach a root cause that points to a specific action.
7. **Fix, verify and reset the reference point.** Make the change, confirm that output improved, and update the standard to the new demonstrated performance.

In the beer filler case, the branch that mattered was how quickly product could be delivered into the container. Following it down led to the diameter of the fill tubes.

A variance tree splits low fills into factors, and one branch holds the cause



Illustrative structure. The bottle is at atmospheric pressure, so there is no back-pressure branch.
The highlighted path is the root cause found in the beer filler case.

variance tree for low fills on a beer filler · 3 branches, 12 factors

The tree shows the factors for a filler that fills at atmospheric pressure. Delivered volume depends on the time available to fill and the rate of flow into the bottle, and the fill target sets what counts as a full bottle. Each leaf is something that can be measured against what it should be. The branch that held the cause is highlighted.

Factors, not brainstorming

A simple analogy shows why the structure matters. Suppose you are twenty minutes late to work every morning. A brainstorm produces a list: traffic, the alarm, the school run, parking, the queue at the coffee shop. Every item is plausible, every item has a champion, and nothing on the list tells you which one matters.

A factor-based approach asks a different question. Your arrival time is made up of three things: when you left home, how long the journey took, and how long it took to get from the car to your desk. Measure each one. If you left on time and the walk was normal, the journey is where the twenty minutes went. Split the journey into the parts of the route and measure those. Within a few steps you know where the time goes, and nobody had to guess.

In a plant, brainstorming has a particular weakness. It returns the usual suspects: the explanations the team already believes, and in a long-running problem, the ones it has already tried. A variance tree forces attention onto parts of the system nobody suspected, because the structure demands that every branch be examined, not just the ones people have theories about. That is how a problem as unglamorous as a tube diameter gets found.

This is not a criticism of experience. Experienced operators and engineers are essential to the method, because they know what "expected" looks like for each factor. The point is that experience on its own pulls you toward familiar causes, and a structured method pulls you toward all of them.

Where this fits in Zero Loss Analysis

Variance Tree Analysis is one tool within the broader Zero Loss Analysis methodology, not the methodology itself. Zero Loss Analysis is the overall discipline of focusing improvement on the constraint, measuring performance against what the equipment can really do, and closing the gap in a structured way. The variance tree is the instrument used at the constraint to decompose the gap into its causes. This paper concentrates on it because it is the tool that most directly unlocks the limits plants believe are permanent.

The size of the opportunity

Across the sites where this methodology has been applied, the pattern is consistent: plants operating at what they believe is 90%-plus efficiency against their own standard are often operating at 60 to 70% of what the equipment can actually deliver, once the standard itself is corrected. That gap — the hidden factory — typically represents a 20 to 40% improvement opportunity that requires no capital investment. The equipment, the labor, and the floor space are all already paid for. The output is simply not being claimed.

The hidden plant

The hidden factory has a bigger sibling, and it shows up on the capital plan rather than the shop floor.

When a manufacturer needs more output, the standard answer is another line. Demand is growing, the existing lines are "maxed out" against the standard, so the business appropriates capital for a second or third line. The business case is built on the same numbers described above: the current ceiling, measured in the accounting system, taken as the true limit of the equipment.

The new line is then built, commissioned and ramped. And more often than anyone would like, it disappoints. It struggles to reach its nameplate rate, and the plant wonders why the promised improvement never materializes. The reason is simple. The new line inherits the same unexamined constraints as the old one, such as the undersized fill tubes, the wrong control logic and the mis-sequenced changeovers. The plant has bought a second copy of the problem.

We call the capacity that was already sitting inside the existing equipment the hidden plant. It is a whole line's worth of output, paid for years ago, that never made it into the capacity plan because the ceiling was written down too low.

This changes the capital conversation. When Zero Loss Analysis recovers capacity at the constraint, the question for the CFO is no longer "where do we find the money for another line?" It becomes "do we still need another line?" In many cases the answer is that the capital can be deferred, and sometimes avoided altogether, because the output the business needs is already available on the equipment it owns.

The practical rule for any capital request that is justified by a capacity shortfall is to test the constraint first. Use Variance Tree Analysis to establish what is really stopping the existing line from running faster, and only then decide whether a new line is needed. If the

constraint is solved, the new line may not be needed. If it is not solved, the new line will simply inherit it.

What this means for continuous improvement teams

The implication for CI teams is direct: before accepting that a constraint has a hard physical limit, ask who decided that, when, and with what method. If the answer is "a team tried a few things a few years ago and moved on," that is not evidence of a physical limit — it is evidence of an unfinished investigation.

The skill that closes this gap is not more brainstorming or more Kaizen events. It is a structured method for decomposing a performance problem into testable factors, and the discipline to keep decomposing until the true cause is isolated — not until the team runs out of obvious guesses.

This is what we teach. Continuous improvement teams that learn Variance Tree Analysis as a structured method, rather than relying on experience-based brainstorming, are consistently able to re-open constraints that were written off as permanent — and recover capacity the plant has already paid for.

Objections you will hear

Teams that start to question their limits meet the same objections again and again. Each has a reasonable answer.

"We have already tried everything." Usually this means everything that occurred to the team. The test is not how many fixes were tried, but whether the problem was ever decomposed so that every factor was examined.

"The equipment vendor says this is the rated speed." A rated speed is a design statement made under the vendor's test conditions. Your product, containers, utilities and configuration are different. A vendor rating is a useful reference point and a poor ceiling.

"It is a quality limit, not a speed limit." The fill tube case is exactly this. A quality defect that appears when speed rises is a symptom with a cause, and the cause can be found.

"We cannot take downtime to investigate." A good deal of the work is observation and measurement during normal production. The disruption from a structured investigation is small compared with the cost of running at the wrong ceiling indefinitely.

"This is probably a one-off." The pattern recurs because the way ceilings get built is the same in every plant: a hard problem, a team that runs out of obvious ideas, and a number that gets written down and defended.

Five questions to ask about any limit

Before accepting that a constraint has reached its physical limit, a continuous improvement team can ask five questions.

1. Who set this limit, when, and on what evidence?
2. What exactly happens when the limit is exceeded, and where in the process does it first show up?
3. Has the problem been broken into measurable factors, or only discussed?
4. Is this limit at the constraint, the one resource that determines the output of the whole plant? A limit elsewhere may be real and still not matter.
5. If the limit were removed, what would it be worth per year, and which capital request would it affect?

If the team cannot answer the first and third questions, the limit is inherited rather than established, and it is a candidate for investigation.

Building the capability: training for continuous improvement teams

The goal of the training is for a plant's own people to find and close these gaps without relying on outside help each time a constraint looks unsolvable. It is built around the team's own constraint, not a generic case study.

The proposed structure has five modules.

1. **Where the real limits are.** Finding the constraint, and mapping which limits are tested and which are inherited.
2. **Reading the gap.** Measuring performance against capability rather than against the standard.
3. **Building a variance tree.** Defining factors, setting expected behavior, and drilling down to root cause.
4. **Running the investigation.** Collecting data at the line, eliminating factors, and verifying the fix.

5. **Making the gain stick.** Resetting standards, reporting the gap, and bringing results into capital planning.

The training can be delivered in two ways: as an on-site workshop with follow-up coaching while your team works a live constraint on its own line, or as a virtual cohort in which teams apply each module to their own line between sessions.

Where to go from here

If your plant's current efficiency targets were set by a team that tried its best and then moved on, there is a reasonable chance your real ceiling is meaningfully higher than your standard says it is.

We run training for continuous improvement teams on Variance Tree Analysis as a practical, structured method — not a theoretical framework — so your own people can find and close these gaps directly, rather than depending on outside consultants every time a constraint looks unsolvable.

If you'd like to talk about what this would look like for your plant, email Trevor at trevor@plantworks.ai or book a 30-minute discovery call.

About the author

Trevor Cusworth is the founder of Plantworks. His career in process manufacturing spans 47 years and includes senior roles at Aspen Technology, Invensys and Schneider Electric, and a period as CEO of Hagen & Company, where the work behind Zero Loss Analysis took shape. In 2009, at Deloitte, he wrote Zero to Hero, the paper that set out the methodology. It was later republished in the Operational Excellence Society Body of Knowledge.